Week 15

Keel Bone Chicken Project

12/08/2025

PURDUE UNIVERSITY | College of Engineering

This Week

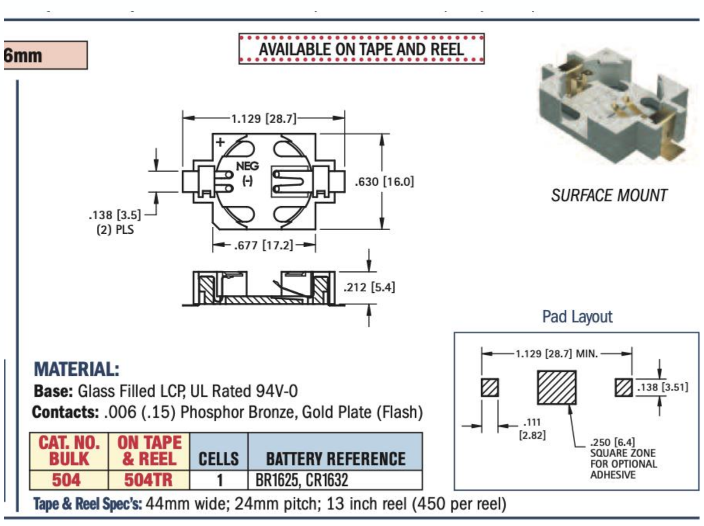
This week's progress

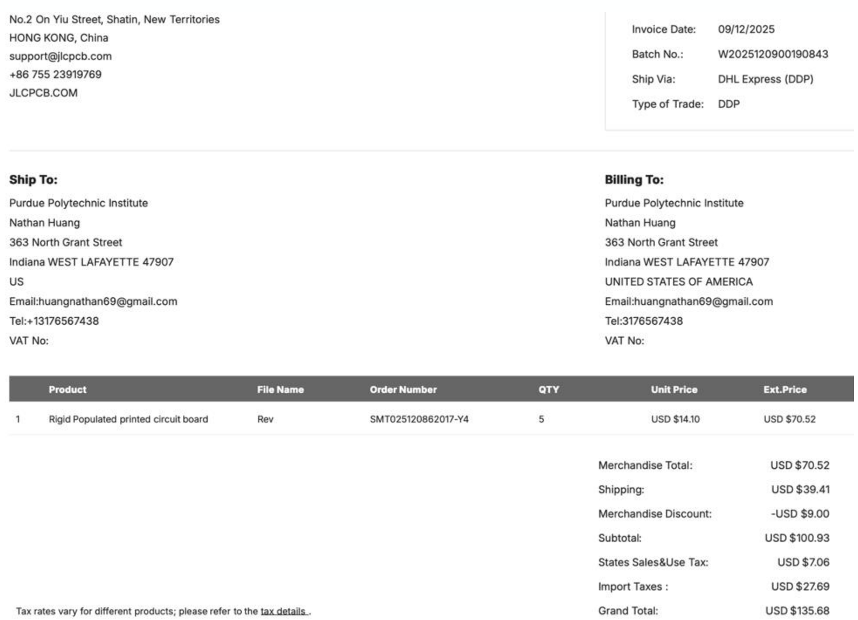
- PCB Invoice
- Selection of PCB Sensors
  - Based on Input from Mechanical Engineering
  - Finalized Sensor Selections with Diagrams
  - Finalized Part Selections with Footprints and Metrics
- Creation of Data Logger Executable
  - Insights into Permanent Solutions for Data Logging Issue

PCB Invoice

PCB Ordered from Procurement Office in POTR

- Effective Shipment Date: **12/11/2025**
- Needed to purchase some equipment on the side (Not available on Vendor website
  - Battery Holder: https://www.digikey.com/en/products/detail/keystone-electronics/504/2745584?utm_campaign=buynow&utm_medium=aggregator&utm_source=snapeda
  - Need to Solder manually





No.2 On Yiu Street, Shatin, New Territories
HONG KONG, China
support@jlcpcb.com
+86 755 23919769
JLCPCB.COM

Invoice Date: 09/12/2025
Batch No.: W2025120900190843
Ship Via: DHL Express (DDP)
Type of Trade: DDP

Ship To:
Purdue Polytechnic Institute
Nathan Huang
363 North Grant Street
Indiana WEST LAFAYETTE 47907
US
Email:huangnathan69@gmail.com
Tel:+13176567438
VAT No:

Billing To:
Purdue Polytechnic Institute
Nathan Huang
363 North Grant Street
Indiana WEST LAFAYETTE 47907
UNITED STATES OF AMERICA
Email:huangnathan69@gmail.com
Tel:3176567438
VAT No:

	Product	File Name	Order Number	QTY	Unit Price	Ext.Price
1	Rigid Populated printed circuit board	Rev	SMT025120862017-Y4	5	USD $14.10	USD $70.52

Merchandise Total:	USD $70.52
Shipping:	USD $39.41
Merchandise Discount:	-USD $9.00
Subtotal:	USD $100.93
States Sales&Use Tax:	USD $7.06
Import Taxes :	USD $27.69
Grand Total:	USD $135.68

Tax rates vary for different products; please refer to the tax details .

Selection of PCB Sensors

Based on Input from Mechanical Engineering

Force Gauge:

- 1.4 m/s Impact Speed Maximum
- 1-2 MPa Contact Pressure on Bone
  - Down from ~10 MPa earlier

Strain Gauge:

- ~5-10% Strain due to Impact
- 1.42 kg approximated for Hen mass

Selection of PCB Sensors

Finalized Sensor Selections with Diagrams

- **Force Sensor** (Capacitive Sensor → Charge Amplifier)
  - Force Sensor: **SingleTact Force Sensor:**
    - https://5361756.fs1.hubspotusercontent-na1.net/hubfs/5361756/SingleTact%20Documents/SingleTact_Datasheet.pdf
  - Charge Amplifier: **AD7150:**
    - https://www.analog.com/en/products/ad7150.html

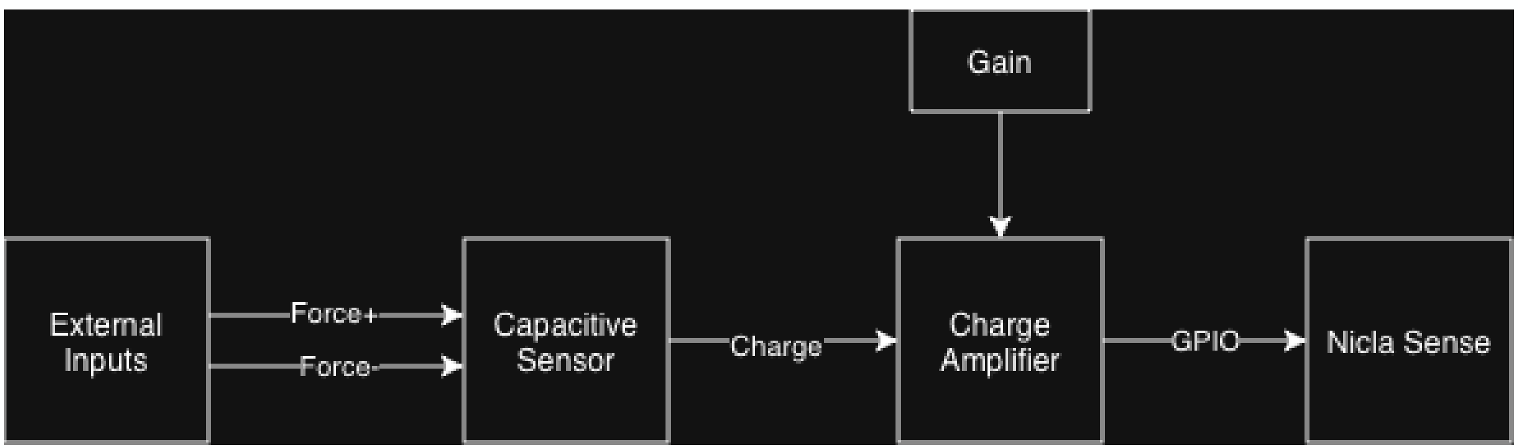
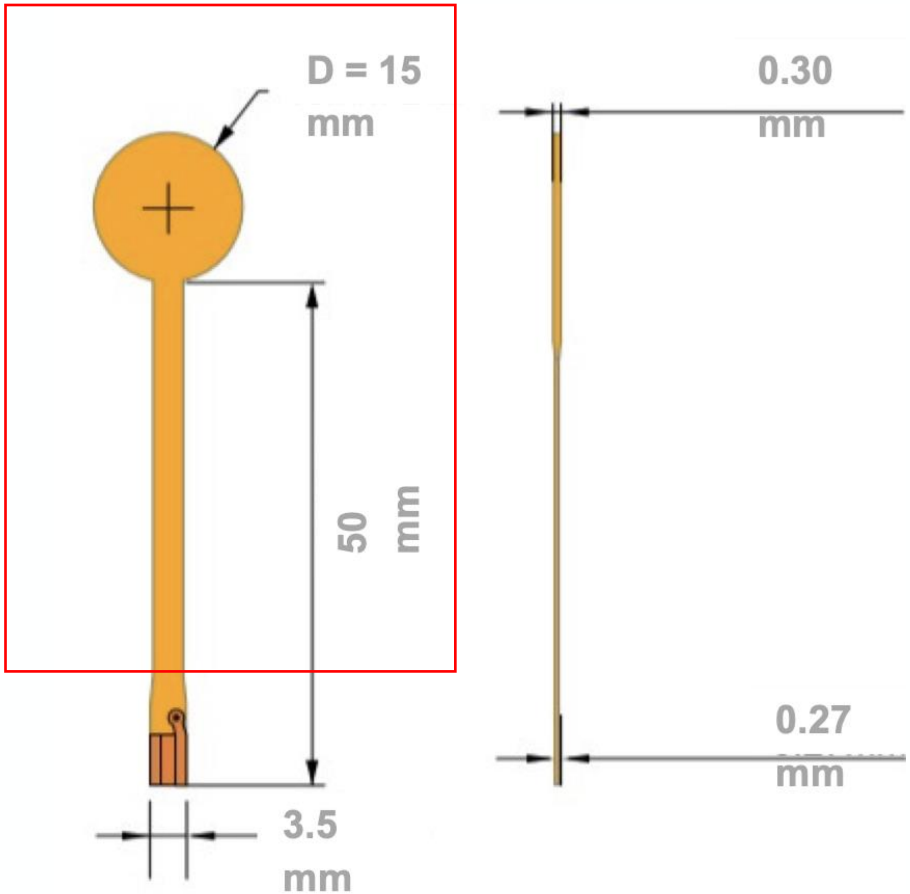


Figure 1: Diagram of Finalized Force Sensor

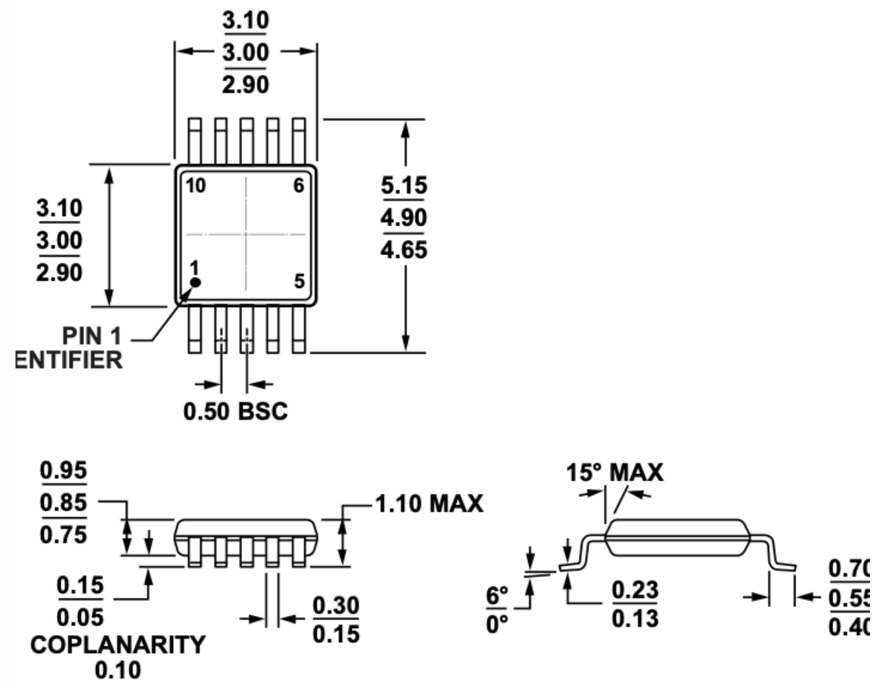
Selection of PCB Sensors

Finalized Part Selections with Footprints and Metrics

Outside region irrelevant



Force Gauge



COMPLIANT TO JEDEC STANDARDS MO-187-BA

Figure 54. 10-Lead Mini Small Outline Package [MSOP] (RM-10)
Dimensions shown in millimeters

Charge Amplifier

Selection of PCB Sensors

Finalized Sensor Selections with Diagrams

- **Strain Sensor** (Resistive Sensor → Wheatstone Bridge)
  - Resistive Sensor: C4A-06-125SLA-120-39P
    - https://mm.digikey.com/Volume0/opasdata/d220001/medias/docus/696/125SL.pdf
  - Charge Amplifier: INA333AIDGKT
    - https://www.digikey.com/en/products/detail/texas-instruments/INA333AIDGKT/1888785?

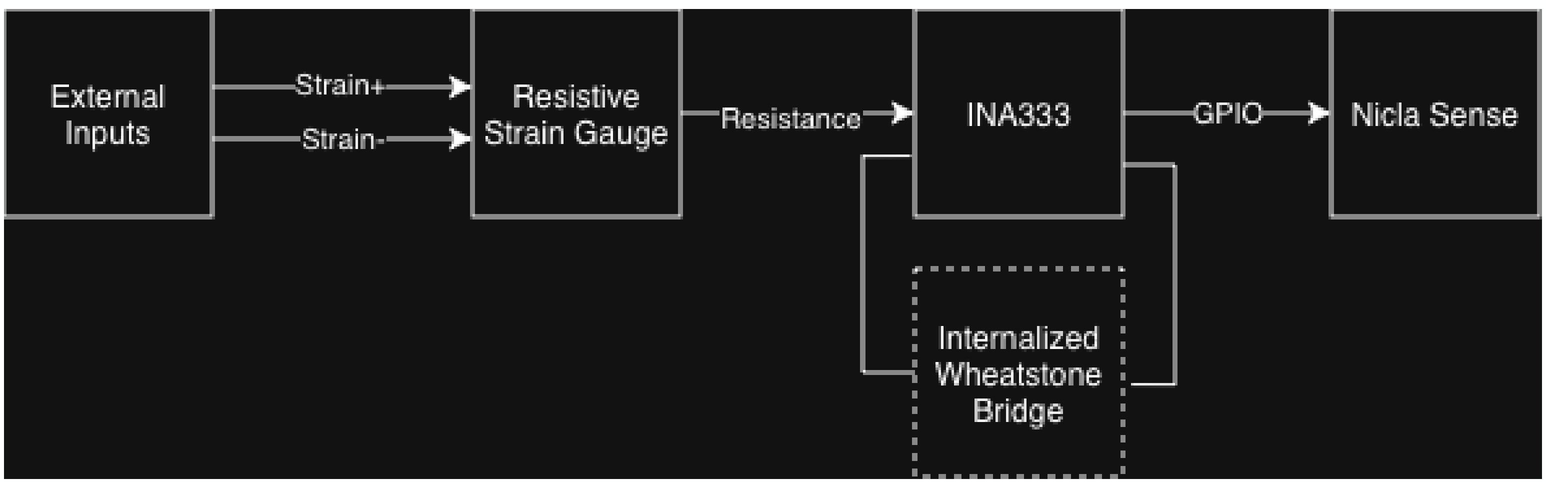


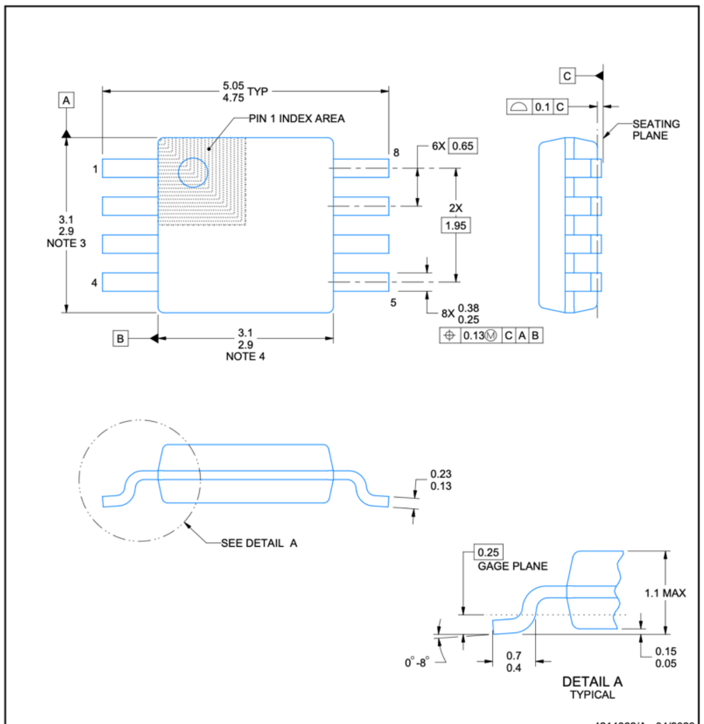
Figure 2: Diagram of Finalized Strain Sensor

Selection of PCB Sensors

Finalized Part Selections with Footprints and Metrics

GAGE DIMENSIONS		Legend ES = Each Section CP = Complete Pattern S = Section (S1 = Section 1) M = Matrix			inch / millimeter
Gage Length	**Overall Length**	**Grid Width**	**Overall Width**	**Matrix Length**	**Matrix Width**
0.125	0.233	0.100	0.130	0.31	0.19
3.18	5.91	2.54	3.30	8.0	4.9

Strain Gauge



NOTES:

PowerPAD is a trademark of Texas Instruments.

Charge Amplifier

Selection of PCB Sensors

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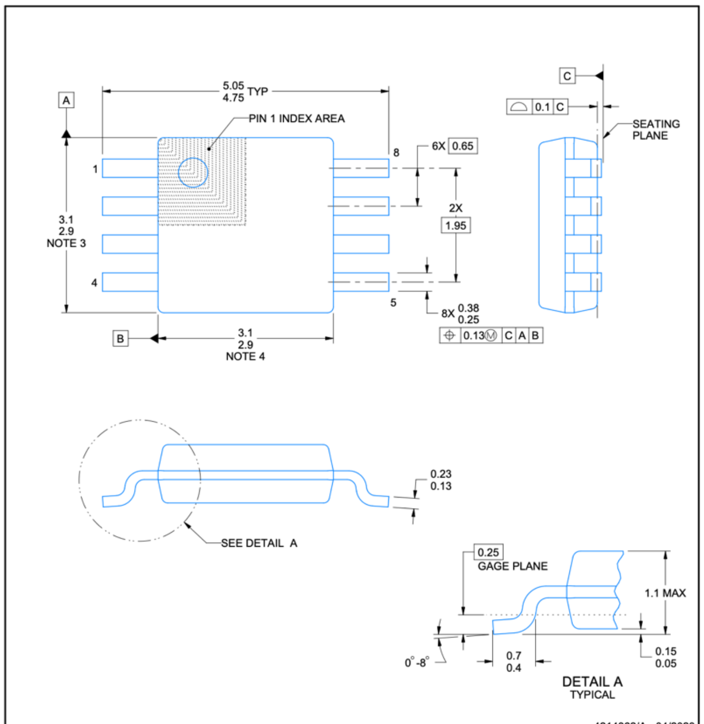

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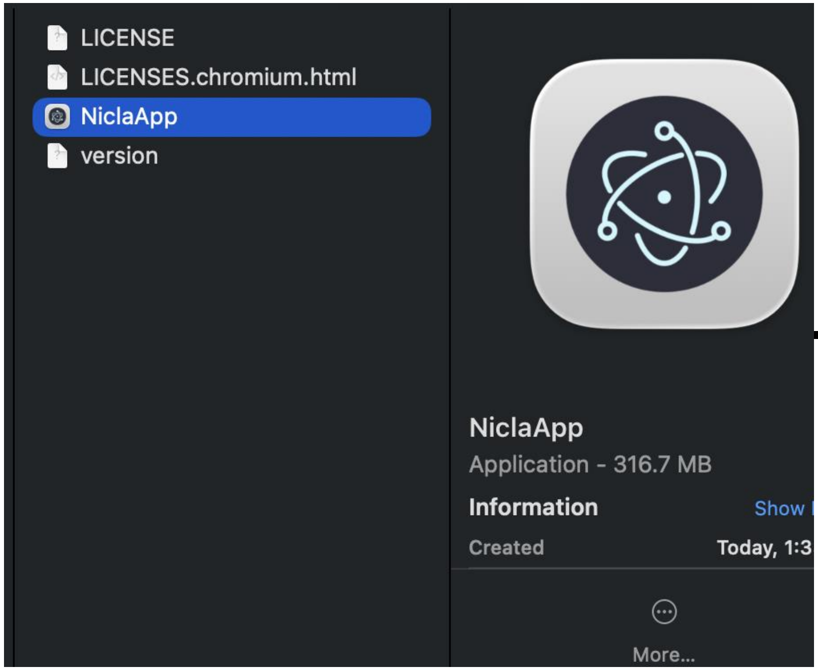
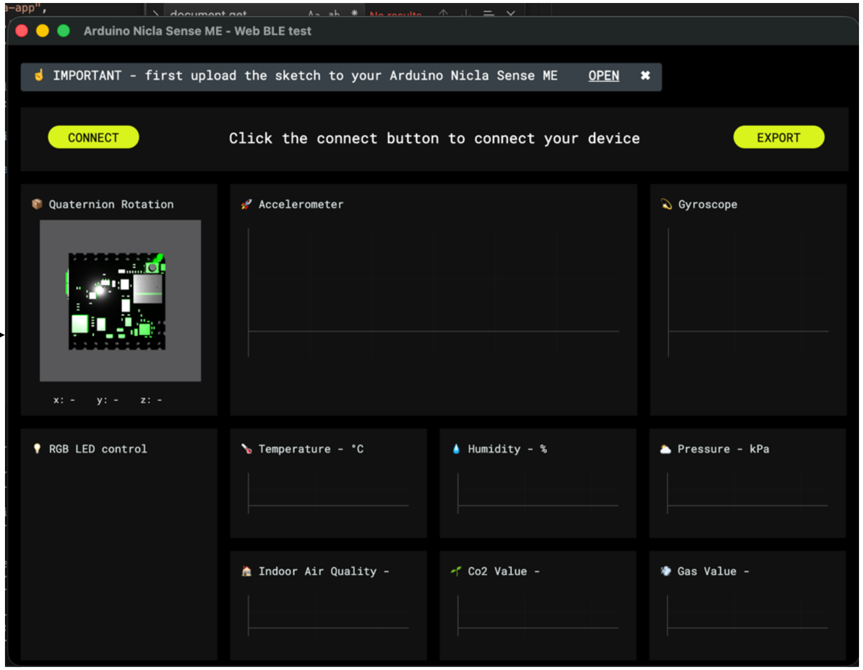
Charge Amplifier

Creation of Data Logger Executable

Creation of Executable to Run on Mac/Windows

- Issues with Data Logging need to be fixed with creation of remote executable
- Tested Hosted Website Compatibility with Omosalewa last Friday





Creation of Data Logger Executable

Creation of Executable to Run on Mac/Windows

- Unsure what options are available with PMDs with Agriculture department:
  - Android/iOS
- Use of embedded iOS development in iOS/Android
  - React Native Frameworks
  - #1 (Expo Go), #2 (NativeBase/Base44)

- **Other Data Storage Options:**
  - Cloud: GitHub (Subscription Required), AWS, Firebase
  - Local: Laptop/Server with SQLite for Database
  - Hybrid: Local Storage + (8 Hour Period x 3) Cloud Backups
  - Executable/App: Installable app logging data locally
    - Local iOS development as seen above

Next Week

Blockers & Next Week's Work

- Hand over PCB to Mechanical + Agriculture Department
  - Write up instructions for use
- Figure out data storage problem with current executable

Blockers:

- Shipment lead time for PCB

Thank You

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